

Problem 5

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Problem 1 True/False: Give an explanation or counterexample. Assume a and L are finite numbers.

- (a) If $\lim_{x \rightarrow a} f(x) = L$, then $f(a) = L$.

Explanation. False. In the graph below $\lim_{x \rightarrow 1} f(x) = 2$, but $f(1)$ does not exist.

alt=graph of function with $f(x)=x+1$, for x not equal to 1. it has an open circle at the point $(1, 2)$

- (b) If $\lim_{x \rightarrow a^-} f(x) = L$, then $\lim_{x \rightarrow a^+} f(x) = L$.

Explanation. False. In the graph below $\lim_{x \rightarrow 2^-} f(x) = -1$ but $\lim_{x \rightarrow 2^+} f(x) = 1$.

alt=graph of a function that is 1 for $x < 2$, but -1 for $x > 2$.

- (c) If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = L$, then $f(a) = g(a)$.

Explanation. False. If we let

alt=graph of a line and parabola, with open circle at their intersection. points are filled in above the intersection for the line and below the intersection for the parabola.

we see that $\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} g(x) = 2$, but $f(1) \neq g(1)$.

- (d) $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ does not exist if $g(a) = 0$.

Explanation. False. If $f(x) = 5x^2$ and $g(x) = x^2$, then $g(0) = 0$ but

$$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 0} \frac{5x^2}{x^2} = \lim_{x \rightarrow 0} 5 = 5.$$